SAHIL GOYAL

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Scholar GitHub LinkedIn Medium

EDUCATION

Indian Institute of Technology Roorkee *2019 - 2024*
B.S.+M.S. in Applied Mathematics
Department Rank 1 (Gold Medal) Overall GPA: **9.44**/10

TECHNICAL SKILLS

Languages and Core Frameworks: Python, C++, JAX (Flax, Optax, Orbax, Chex), PyTorch

PUBLICATIONS

1. **ELT: Elastic Looped Transformers for Visual Generation.** [Paper]
   Sahil Goyal*, Swayam Agrawal*, Gautham Anil, Prateek Jain, Sujoy Paul, Aditya Kusupati [**ECCV** '26]
2. **MaGNeTS: Masked Generative Nested Transformers with Decode Time Scaling.** [Paper] [Poster]
   Sahil Goyal*, Debapriya Tula*, Gagan Jain, Pradeep Shenoy, Prateek Jain, Sujoy Paul [**ICML** '25]
3. **Design-o-meter: Towards Evaluating and Refining Graphic Designs.** [Paper] [Poster] [Survey]
   Sahil Goyal*, Abhinav Mahajan*, Swasti Mishra, Prateksha Udhayanan, Tripti Shukla, K J Joseph, Balaji Vasan Srinivasan. [**WACV** '25]
4. **Emotionally Enhanced Talking Face Generation.** [Paper] [Poster] [Code]
   Sahil Goyal, Shagun Uppal, Sarthak Bhagat, Yi Yu, Yifang Yin, Rajiv Ratn Shah. [**ICCV-W '23**] [**ACM MM-W '23**]
5. **Emotional talking faces: Making videos more expressive and realistic.** [Paper] [Demo]
   Sahil Goyal, Shagun Uppal, Sarthak Bhagat, Dhroov Goel, Sakshat Mali, Yi Yu, Yifang Yin, Rajiv Ratn Shah. [**ACM MM Asia '22**] [**Best Demo paper**]

EXPERIENCE

***Pre-Doctoral Researcher* at Google DeepMind** *June 2024 - July 2026*
Advisors: Dr. Prateek Jain, Dr. Sujoy Paul, Dr. Aditya Kusupati

- Research in **Efficient ML** (training-time compute efficiency, inference-time compute efficiency, and parameter efficiency) including contributions to Veo and Gemini, and publications at ICML [2] and ECCV [1].

***Research Intern* at Adobe Research** *May 2023 - July 2023*
Advisors: Dr. Joseph KJ, Dr. Balaji Vasan Srinivasan

- Unified framework [3] for automated design evaluation and refinement, enabling iterative improvements for creators.

***Research Intern* at MIDAS Lab @IIITD** *Sep 2021 - Jan 2023*
Advisors: Dr. Rajiv Ratn Shah

- Multimodal talking face generation network [4, 5], winning **Best Demo Paper** award at ACM MM Asia.

***Machine Learning Intern* at PANDO** *March 2021 - April 2021*

- Developed a **20x** faster 3-D space optimization algorithm for transport logistics.

KEY RESEARCH PROJECTS

Elastic Looped Transformers for Visual Generation (ECCV [1]) *Oct 2025 - April 2026*

- Improved discrete and continuous diffusion models with **4x** less parameters (FID of **2.0** using just ∼**100M** params).
- Developed novel **Intra-Loop Self-Distillation**, enabling **Elastic Inference** and dynamic compute scaling.
- Achieved **3.5x** inference throughput and ∼**35%** lower peak HBM compared to standard transformer baselines.

Compression of Pre-trained LLMs (Gemini) *May 2025 - July 2026*

- Optimized **model compression** pipeline for Gemini, aiming for **minimal extra training cost**.
- Built novel and effective **structured-pruning** and **layer-wise distillation** techniques.
- This involved **pre-training**, **finetuning** and **evaluation** of Gemini models across multiple scale tiers.
- Results show significant quality gains over iso-params and iso-FLOPs baselines.

Adaptive Compute for Visual Generation (Veo, ICML [2]) *June 2024 - Feb 2025*

- Developed an **adaptive computation** method [2] for multi-step generative models using **nested transformers**.
- Achieved ∼**3x** reduction in inference FLOPs for image/video generation.
- Implemented similar efficient techniques for Google's SOTA video generation model (Veo).
- Executed **end-to-end training** and **evaluation** of Veo models.

Automated Evaluation and Refinement of Graphic Designs (WACV [3]) *May 2023 - July 2023*

- Built compact **Siamese** networks (**0.5M** params) to score Graphic Designs.
- Significantly outperformed GPT-4o and LLaVA-NeXT (7B) in ranking graphic designs.
- Tailored advanced **Genetic Algorithms** for SOTA **Layout refinement** (mean-IOU of **0.54** on Crello dataset).

Emotionally Enhanced Talking Face Generation (ICCV-W [4], Best Demo paper [5]) *Sep 2021 - Jan 2023*

- Developed the first **talking face generation** network [4, 5] that incorporated emotions.
- SOTA results on **CREMA-D** dataset (**83.2**% emotion classification accuracy and **FID** metric: **5.29**).
- Deployed the entire system as an interactive web interface.

SELECTED ACHIEVEMENTS

- **PhD admits** from UT Austin, EPFL, MILA, ELLIS (Declined). *2026*
- **2 Patents** filed at US Patent Office on Efficient Visual Generation [Patent 1]. *2025*
- Among **15**, from 18,000+ applicants, selected for Google's Pre-Doctoral Researcher Program. *2024*
- **Department Gold Medal** (IIT Roorkee - Applied Mathematics). *2024*
- **Best Demo Paper** at ACM MM ASIA conference. *2022*
- **Gold Medal** at Inter IIT Tech Meet. *2022*
- **Open Source**: PyTorch-Vision (debugged dataloaders, ported tests and updated docs). *2022*
- Awarded ACM SigMM student travel grant for ACM MM Asia. *2022*
- JEE Advanced Rank of **1583** among 1.5M students. *2019*
- **600**+ stars on GitHub.

POSITIONS OF RESPONSIBILITY

- **Reviewer** at NeurIPS, ICML, ICLR, CVPR, ECCV, ICCV, AAMAS, WACV conferences. *2024 - Present*
- **Mentor** at Data Science Group IITR *2020 - 2024*
- **Undergraduate Teaching Assistant** at Academic Reinforcement Program IITR *2021*